EQUATION, RATION, PROPORTION AND VARIATION

Directions for questions 1 to 4: Select the correct alternative from the given choices.

A profit of Rs.500 is divided between A and B in the ratio 1/2 : 1/3. What will be the difference in their profit shares?

- **a)** Rs.200
- **b)** Rs.100 (Your answer - ✅)
- **c)** Rs.300
- **d)** Rs.50

Solution:

The shares of A, B and their difference are listed below.

Difference	A	B	Sum
	$\frac{1}{2}$	$\frac{1}{3}$	
1	3	2	5
100			500

The shares of A, B are 300 and 200. The difference is 100. Choice (B)

Directions for questions 1 to 4: Select the correct alternative from the given choices.

In a family, the father took $1/5^{th}$ of a cake and the rest of the cake was distributed equally among all the others in the family. If the father had two times as much cake as each of the others had, the total number of family members is ______.

- **a)** 16
- **b)** 15
- **c)** 8
- **d)** 9

Solution:

The father had $\frac{1}{5}$ and the remaining $\frac{4}{5}$ was equally distributed. As each of the others had half what the father had, each had $\frac{1}{10}$. The number of the other members is $\frac{4}{5} / \frac{1}{10}$, i.e., 8. The total number of members is 1 + 8, i.e., 9. Choice (D)

Directions for questions 1 to 4: Select the correct alternative from the given choices.

The third proportional to

$\frac{x}{y}+\frac{y}{x}$ and $\frac{x}{y}$ is

______.

a) $\frac{(x+y)^2}{xy}$

b) $\frac{x^2y}{y(x^2+y^2)}$

c) $\frac{x^3y}{x^2+y^2}$

d) $\frac{x^3}{y(x^2+y^2)}$

Solution:

The third proportional to a, b is b^2/a

$\therefore$ The third proportional to $\frac{x}{y}+\frac{y}{x}$ and $\frac{x}{y}$ is $\frac{x^2}{y^2} \times \frac{xy}{x^2+y^2} = \frac{x^3}{y(x^2+y^2)}$

Choice (D)

Directions for questions 1 to 4: Select the correct alternative from the given choices.

At constant temperature the volume of a gas varies inversely as its pressure. When the pressure of a certain gas is 80 units, its volume is 45 units. If the pressure is 25 units, what is its volume?

- **a)** 144 units
- **b)** 15 units
- **c)** 90 units
- **d)** None of these

Solution:

The pressure and volume are inversely proportional. The data and conclusion are listed below.

P	V	PV
80	45	16(9) (25)
25	144	16(9) (25)

The required volume would be 144 units Choice (A)

Directions for question 5: Type in your answer in the input box provided in the question.

A two-digit number is four times the sum of its digits. When 36 is added to the number, the digits are reversed. The number is

Solution:

Let the two digit number be 'xy'.

$10x + y = 4x + 4y \Rightarrow 2x = y$

Also $10x + y + 36 = 10y + x \Rightarrow x + 4 = y$

$\therefore$ 'xy' = 48. Ans (48)

Directions for question 6: Select the correct alternative from the given choices.
The sum of the present ages of X, Y and Z is 45 years. Three years ago their ages were in the ratio 1 : 2 : 3. What is the present age of X?

- **a)** 10 years
- **b)** 6 years
- **c)** 8 years
- **d)** 9 years

Solution:

The sum of the present ages of X, Y, Z is 45 years. Three years ago, the sum was 36 years. As their ages at that time were in the ratio of 1 : 2 : 3, they were 6, 12, and 18 respectively. X's present age is 9. Choice (D)

Directions for question 7: Type in your answer in the input box provided below the question
The present age of a grandfather is 5 more than ten times the age of his grandson. After how many years will the grandson attain the present age of the grandfather, which is65 years?

Solution:

Let the grandson's age be x. The man's age is 10x + 5. $10x + 5 = 65 \Rightarrow x = 6$. The grandson will attain the age of 65 after 59 years. Ans: (59)

Directions for questions 8 to 12: Select the correct alternative from the given choices.

If $7x - 3y = 25$ and $6x + 6y = 90$, what are the values of *x* and *y*?

- **a)** 6, 7
- **b)** 7, 8
- **c)** 9, 10
- **d)** 5, 8

Solution:

$7x - 3y = 25 \Rightarrow 14x - 6y = 50$ .... (1)

$6x + 6y = 90$ ....(2)

$(1) + (2) \Rightarrow 20x = 140 \Rightarrow x = 7. \therefore y = 8.$

Choice (B)

Directions for questions 8 to 12: Select the correct alternative from the given choices.

The voltage across a wire is directly proportional to the current passing through it. In a particular instance, when voltage is 36 volts, the current passing through the wire is 54 amperes. At what voltage will the current passing through the wire be 90 amperes?

- **a)** 72 volts
- **b)** 90 volts
- **c)** 60 volts
- **d)** 54 volts

Solution:

When the current is 54A, the voltage is 36 V.

$\therefore$ when the current is 90A, the voltage would be $\frac{90}{54}(36)$ V i.e., 60 V.

Choice (C)

Directions for questions 8 to 12: Select the correct alternative from the given choices.

Fifteen oranges and seven apples cost as much as ten oranges and nine apples. What is the ratio of the cost of one orange to that of one apple?

- **a)** 2 : 5
- **b)** 5 : 2
- **c)** 3 : 4
- **d)** Cannot be determined

Solution:

Let the cost of an orange be x and an apple be y.

$15x + 7y = 10x + 9y \Rightarrow 5x = 2y \Rightarrow \frac{x}{y} = \frac{2}{5}$

Choice (A)

Directions for questions 8 to 12: Select the correct alternative from the given choices.

If Rs.275 is divided into three parts such that half of the first part, one-fourth of the second part and one-fifth of the third part, are all equal. The three parts are ______.

- **a)** Rs.20, Rs.125, Rs.130
- **b)** Rs.25, Rs.100, Rs.150
- **c)** Rs.100, Rs.50, Rs.25
- **d)** Rs.50, Rs.100, Rs.125

Solution:

The parts are 2x, 4x and 5x. As 11x = 275, x = 25. The parts are 50, 100, and 125.
Choice (D)

Directions for questions 8 to 12: Select the correct alternative from the given choices.

A varies inversely with B. When B is 64 units, A is 36 units. Find A when B is 48 units.

- **a)** 54 units
- **b)** 94 units
- **c)** 48 units (Your answer - ✅)
- **d)** 72 units

Solution:

$A \propto \frac{1}{B}$, i.e., AB = constant.

A	B	AB
$36 = 2^2 3^2$	$64 = 2^6$	$2^8 3^2$
$2^4 3^1$	$48 = 3(2^4)$	$2^8 3^2$

When B = 48, A has to be 48, so that the product AB remains constant. Choice (C)

Directions for question 13: Type in your answer in the input box provided below the question

In a cricket test series the runs made by Mark and Steve are in the ratio 5 : 9 and that of Hayden and Steve are in ratio 6 : 7. If Hayden scores 162 runs, then what are the runs made by Mark?

Solution:

The data (and the conclusion) are shown below.

Mark	Steve	Hayden
5	9	-
-	7	6
35	63	54
105		162

Mark made 105 runs Ans : (105)

Directions for question 14: Select the correct alternative from the given choices.

Sum of the numerator and the denominator of a fraction is 22. If two is added to the numerator and three is added to the denominator, the fraction becomes 1/2. The fraction is

a) $\frac{3}{7}$

b) $\frac{7}{16}$

c) $\frac{7}{15}$ (Your answer - ☑)

d) $\frac{6}{11}$

Solution:

Let the fraction be $\frac{N}{22-N}$.

$\frac{N+2}{25-N} = \frac{1}{2} \Rightarrow 2N + 4 = 25 - N \Rightarrow N = 7$

The fraction is $\frac{7}{15}$. Choice (C)

Directions for question 15: Type in your answer in the input box provided below the question

In a group of cows and hens the total number of legs is 10 more than thrice the total number of heads. If the number of cows and hens put together is 30, then find the number of cows.

Solution:

Let the number of cows and hens be x and y respectively. $4x + 2y = 10 + 3(x + y)$ $\Rightarrow x - y = 10$. Also $x + y = 30 \therefore x = 20, y = 10$. The number of cows is 20.

Ans : (20)

Directions for questions 16 to 18: Select the correct alternative from the given choices.

The ratio of a person's age to his son's age is 5 : 2. If after ten years the age of the son is half of his father's age, the ratio of their ages after four years will be _______.

- **a)** 11 : 6
- **b)** 8 : 3
- **c)** 9 : 4 (Your answer - ✅)
- **d)** 5 : 6

Solution:

Let the son's ages be 2x. The father's age is 5x. Ten years hence, the age will be 2x + 10, 5x + 10 respectively. $\therefore 2(2x + 10) = 5x + 10 \Rightarrow x = 10$.
Their ages are 20, 50 respectively. Four years hence, they will be 24, 54. The ratio of the father's age and son's age, 4 years hence, is 54 : 24 = 9 : 4 Choice (C)

Directions for questions 16 to 18: Select the correct alternative from the given choices.

Rs.1,344 is divided into three parts which are proportional to 3, 5 and 4. Find the difference between the largest amount and the smallest amount.

- **a)** Rs.336
- **b)** Rs.560
- **c)** Rs.224 (Your answer - ✅)
- **d)** Rs.300

Solution:

The 3 parts, (say a, b, c, where a < b < c), the difference of the largest and the smallest, the sum of the 3 parts are listed below.

c – a	a	b	c	a + b + c
2	3	4	5	12
2 (112)				1344

∴ The required difference is 224. Choice (C)

Directions for questions 16 to 18: Select the correct alternative from the given choices.

Ten years hence, Srinivas will be twice as old as his son. But two years ago, he was four times as old as his son. Find the present age of his son.

- **a)** 10 years
- **b)** 6 years
- **c)** 12 years
- **d)** 8 years

Solution:

The data is tabulated below.

	P – 2	P + 10
Srinivas	4x	4x + 12
Son	x	x + 12

$\therefore$ $4x + 12 = 2(x + 12) = 2x + 24 \Rightarrow x = 6$,
The son's present age is $x + 2$, i.e. 8.

Choice (D)

Directions for question 19: Type in your answer in the input box provided below the question

A house has provisions sufficient for four men to last for 30 days. If an additional relative stays with them, then for how many days will the provisions last?

Your Answer: 24 ✅

Directions for question 20: Select the correct alternative from the given choices.

There are two numbers such that the sum of twice the first and thrice the second is 23. The sum of four times the first and five times the second is 41. The difference between the two numbers is ______.

- a) 4
- b) 5
- c) 1 (Your answer - ✅)
- d) 2